

Algorithmus and the Final Project

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Bob is taking the notoriously difficult Advanced Algorithms course this semester, taught by the legendary, eccentric Professor Algorithmus.

Professor Algorithmus has devised a highly complex skill formula to calculate a precise “Skill Rating“ for each of his n students, where student i is assigned a skill rating a_i .

It is time for the final project, and Professor Algorithmus needs to divide his class of n students into project teams. A team can consist of any number of students, from just a single person up to the entire class.

However, he knows that if the skill difference between members of the same team is too big, the student with a high skill rating will end up doing all the work, while the student with a low skill rating will end up learning nothing. So, he made one strict rule: for any given team, the difference between the maximum Skill Rating and the minimum Skill Rating of the students in that team must be less than or equal to k .

Professor Algorithmus is extremely lazy and hates grading. To minimize his own workload, he asks you for help. Given the skill rating of each student and an integer k , can you find the minimum possible number of teams he can create while respecting the rule?

Input

The first line contains two integers n and k ($1 \leq n \leq 1000$; $0 \leq k \leq 10^9$) — the number of students in the class and the maximum allowed Skill Rating difference within a single team.

The second line contains n space-separated integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$) — the calculated Skill Rating of each student.

Output

Output a single integer — the minimum number of project teams Professor Algorithmus can form.

Examples

standard input	standard output
5 2 1 5 3 2 6	2
4 0 10 10 10 10	1